



Development of Highly Selective Support for CO Hydrogenation to Light Olefins with Partially Passivated Iron Catalysts

Yu Xing,^{*,[a]} Gaopeng Jia,^[a] Zhenxin Liu,^{*,[a]} Shaoming Fang,^[a] Chenxi Zhao,^[a] Xuehui Guo,^[a] and Steven L. Suib^[b]

To rule out disturbances from water-soluble efficient promoters (e.g., K_2O or combination of S/Na), a simplified iron/support catalyst system was employed to investigate novel supports for the catalytic conversion of syngas ($CO + H_2$) to light olefins (FTO). A series of supports with nominal compositions of $ZnAl_2O_4 \cdot xAl_2O_3$ ($x = 1, 2, 3$) were prepared by coprecipitation and subsequent high temperature calcination at $1200^\circ C$ for 24 h to generate partially passivated $ZnAl_2O_4$ and Al_2O_3 phases, through which each formed component of a catalyst may minimize interactions among them and exhibit unique activity. Iron catalysts were prepared by incipient wetness of iron salts

over these supports. At a total pressure of ca. 20 bar and a CO conversion of 35 %, a catalyst with a nominal composition of 15%Fe/85%($ZnAl_2O_4 \cdot 3Al_2O_3$) exhibits a surprising $C_2=C_4$ hydrocarbon distribution of 58 %, which reaches a huge increase by almost 100 % in comparison with iron catalysts over regular supports (e.g., carbon, gamma- Al_2O_3 , etc.). This catalyst demonstrates the $1200^\circ C$ passivated $ZnAl_2O_4 \cdot 3Al_2O_3$ as probably the most selective support material for iron-based FTO catalysts. At CO conversion of 35 %, the contents of ethylene in C_2 , propylene in C_3 , and butylene in C_4 products achieve extraordinary values of 82 %, 92 %, and 91 %, respectively.

1. Introduction

Conversion of biomass/coal-based synthesis gas (i.e., syngas) or CO_2 -containing syngas may be applied to produce long chain hydrocarbons in a process called Fischer-Tropsch synthesis (FTS),^[1–4] or light olefins in a process called Fischer-Tropsch to olefins (FTO, also called FTTO),^[5–9] or even oxygenates such as alcohols.^[10–12] Light olefins (i.e., $C_2=C_4$) contain two to four carbon atoms. The literature about direct and indirect FTO processes has been well summarized.^[5]

Three major routes, i.e., oxide/zeolite catalysts, cobalt-based catalysts, and iron-based catalysts, have been investigated for direct FTO process with several milestone-level reports by journals like *Science* and *Nature*.^[6–9,13] Typical oxide/zeolite methods are unique metal-free routes developed by F. Jiao and X. Bao et al. (in 2016) using $ZnCrO_x/MSAPO-34$ ^[8] as well as by K. Cheng and Y. Wang et al. (in 2016) using $ZrO_x-ZnO/SAPO-34$,^[13] which more exactly belong to the tandem catalysis that couples the methanol synthesis and the C–C coupling reaction and may lead to very high $C_2=C_4$ hydrocarbon distribution values of

80%^[8] and 74%.^[13] Typical cobalt-based catalyst method of L. Zhong and Y. Sun et al. (in 2016) about cobalt carbide nanoprisms^[9] is very good for generating light olefins at low temperature and low total pressure. Typical iron-based catalyst methods of H.M. Torres Galvis and K.P. de Jong et al. (in 2012), combining S and Na elements as promoters with synergy effects to Fe/CNF and Fe/ $\alpha-Al_2O_3$,^[6,7] reach high $C_2=C_4$ hydrocarbon distribution value of 53 % at a total pressure of 20 bar (this value is 64 % at a total pressure of 1 bar), which may have potential problems due to the encounter of sulfur impurities in the syngas feedstock and, as a result, probable change of S content in the catalyst.

In comparison with simple oxides (e.g., Al_2O_3 or SiO_2), complex oxides (e.g., AB_2O_4 spinels or ABO_3 perovskites) usually have higher resistance against steam and thermal sintering.^[14,15] Although iron itself has good water-gas shift (WGS) activity and will not be seriously destroyed by steam conditions, the simple oxide support phase of an iron-based catalyst may have to suffer from steam environment and formation of unfavorable metal-support compounds.^[16] The development of complex oxides supported iron (or cobalt) nanocomposites may be promising to better fit the steam conditions of FTO and FTS.^[16–18] Selectivity modulation of CO hydrogenation by special element attracts great attentions.^[19–29] Since the beginning of 2015, we have demonstrated that spinel-type supported cobalt catalysts are tolerant against sulfur poisons and steam during CO hydrogenation.^[16–18,30] Literature about spinel-supported iron-based catalysts for CO hydrogenation to light olefins is still sparse.^[31] Herein we report our investigations on novel high temperature passivated spinel/alumina composite supports in a simplified iron/support system, which may rule out disturbances from water-soluble efficient promoters (e.g., K or combination

[a] Prof. Dr. Y. Xing, G. Jia, Prof. Dr. Z. Liu, Prof. Dr. S. Fang, C. Zhao, X. Guo
Henan Provincial Key Laboratory of Surface and Interface Science
School of Materials and Chemical Engineering
Zhengzhou University of Light Industry
Zhengzhou, 450002 (P.R. China)
E-mail: yuxing@zzuli.edu.cn
zhenxinliu@zzuli.edu.cn

[b] Prof. Dr. S. L. Suib
Department of Chemistry, and Department of Chemical
Materials & Biomolecular Engineering
University of Connecticut
Storrs, CT 06269-3060 (USA)

Supporting information for this article is available on the WWW under <https://doi.org/10.1002/cctc.201900023>

Table 1. Nomenclature and crystalline phases of samples.^[a]

sample code	nominal composition (metal/support) (wt)	x value ^[b]	crystalline phases in XRD patterns of supports or freshly reduced catalysts	crystalline phases in XRD patterns of used catalysts
F1	15 %Fe/85 % (ZnAl ₂ O ₄ · 1Al ₂ O ₃)	1	Fe ⁰ (major), ZnAl ₂ O ₄ (medium), θ-Al ₂ O ₃ (medium)	ZnAl ₂ O ₄ (major), θ-Al ₂ O ₃ (medium)
F2	15 %Fe/85 % (ZnAl ₂ O ₄ · 2Al ₂ O ₃)	2	Fe ⁰ (major), ZnAl ₂ O ₄ (minor), θ-Al ₂ O ₃ (minor), α-Al ₂ O ₃ (few)	ZnAl ₂ O ₄ (major), θ-Al ₂ O ₃ (major), α-Al ₂ O ₃ (minor), β-Al ₂ O ₃ (minor)
F3	15 %Fe/85 % (ZnAl ₂ O ₄ · 3Al ₂ O ₃)	3	Fe ⁰ (major), ZnAl ₂ O ₄ (medium), θ-Al ₂ O ₃ (medium), α-Al ₂ O ₃ (medium)	ZnAl ₂ O ₄ (major), θ-Al ₂ O ₃ (major), α-Al ₂ O ₃ (major), β-Al ₂ O ₃ (minor)
F _∞	15 %Fe/85 %Al ₂ O ₃	∞	Fe ⁰ (major), β-Al ₂ O ₃ (major), α-Al ₂ O ₃ (minor)	β-Al ₂ O ₃ (major), α-Al ₂ O ₃ (minor)
REF-a	15 %Fe/85 % (ZnAl ₂ O ₄ · 1Al ₂ O ₃) ^[c]	1	Fe ⁰ (major), ZnAl ₂ O ₄ (major)	ZnAl ₂ O ₄
REF-b	100 %Fe ^[c]	N/A	Fe ⁰	Fe ₃ C ₂
SP-0	100 %ZnAl ₂ O ₄	0	ZnAl ₂ O ₄	N/A
SP-1	100 % (ZnAl ₂ O ₄ · 1Al ₂ O ₃)	1	ZnAl ₂ O ₄ (major), θ-Al ₂ O ₃ (medium), β-Al ₂ O ₃ (few)	N/A
SP-2	100 % (ZnAl ₂ O ₄ · 2Al ₂ O ₃)	2	ZnAl ₂ O ₄ (major), θ-Al ₂ O ₃ (major), β-Al ₂ O ₃ (medium), α-Al ₂ O ₃ (minor)	N/A
SP-3	100 % (ZnAl ₂ O ₄ · 3Al ₂ O ₃)	3	ZnAl ₂ O ₄ (major), α-Al ₂ O ₃ (major), θ-Al ₂ O ₃ (major), β-Al ₂ O ₃ (minor)	N/A
SP-∞	100 %Al ₂ O ₃	∞	β-Al ₂ O ₃ (major), α-Al ₂ O ₃ (minor)	N/A

[a] F1, F2, F3, F_∞, REF-a and REF-b are “catalyst” samples. SP-0, SP-1, SP-2, SP-3 and SP-∞ are “support” samples. [b] $x = n(\text{Al}_2\text{O}_3)/n(\text{ZnAl}_2\text{O}_4)$, i.e., molar ratio of nominal Al₂O₃ to nominal ZnAl₂O₄. [c] The calcination temperatures for REF-a and REF-b are both 350 °C.

of S/Na). K.P. de Jong et al. made a milestone on highly efficient promoter for iron FTO catalyst, while we report a breakthrough on highly selective support material for iron FTO catalyst.

Experimental Section

Preparation of Materials

The sources for Zn and Al elements are Zn(NO₃)₂·6H₂O and Al(NO₃)₃·9H₂O respectively, while their dosages are both based on the ratios of elemental compositions shown in Table 1. Supports (SP-0~SP-∞) are prepared in the sequence of coprecipitation, extrusion, drying, and subsequent calcination at 1200 °C for 24 h. For example, support SP-1 is synthesized with a precipitant (i.e., sodium bicarbonate) and a solution of Zn/Al sources at 40 °C under vigorous stirring via conventional coprecipitation, filtered after one hour digestion, thoroughly washed with water, and then partially desiccated at 80 °C. The half-dried filter cake was mixed with sesbania gum powder (i.e., an agent for assisting extrusion) and extruded, air-dried at RT, desiccated at 110 °C for 2 h, and then calcined in a tubular furnace (i.e., equipped with a normal alundum tube) at 1200 °C for 24 h in air atmosphere. The received support material is denoted as SP-1. Other support materials are prepared similarly. Unreduced catalysts F1, F2 and F3 are respectively prepared by incipient wetness of supports SP-1, SP-2, and SP-3 with aqueous solutions of ammonium iron citrate (C₆H₁₁FeNO₇), followed by calcination at 350 °C for 4 h in air atmosphere.

Unreduced reference catalyst REF-a is prepared with the precipitant (i.e., sodium bicarbonate) and a solution of Fe/Zn/Al sources, of which the source of Fe element is Fe(NO₃)₃·9H₂O. Unreduced reference catalyst REF-b is prepared with the precipitant (i.e., sodium bicarbonate) and a solution of Fe(NO₃)₃·9H₂O. After the procedures of coprecipitation, extrusion and drying, subsequent calcination for both catalyst REF-a and catalyst REF-b are conducted at 350 °C in air atmosphere. More details for the preparation of each material are provided in Supporting Information.

Characterization

Crystalline phases are identified by powder X-ray diffraction (XRD) using a Panalytical X'Pert Pro diffractometer with Cu K α radiation. Field-emission scanning electron microscopy (FESEM) studies are carried out using a JSM-7001F FESEM. Elemental contents are measured with an Agilent 725 ICP-OES. A Quantachrome NOVA1000 surface area and pore size analyzer is used for nitrogen sorption measurements of mesoporosity. Ex-situ X-ray photoelectron spectroscopy (XPS) is performed on a Thermo Fisher Scientific Escalab 250XI spectrometer with a monochromated Al_{K α} excitation source without using ion etching. After a FTO reaction, the reactor is cooled down and purged with ultra-high-purity 100 %Ar at RT. Then the used catalyst is discharged at RT from the reactor and transferred in air into the XPS instrument. The air transfer procedure is employed because our XPS instrument is not equipped with a controlled atmosphere reaction chamber. The C1s is adjusted to 284.8 eV as a reference signal. The curve fitting of XPS data is conducted using Avantage 5.52 Surface Chemical Analysis software. CO₂ temperature-programmed desorption (CO₂-TPD) is performed using a Quantachrome ChemStar™ chemisorption analyzer (Detector: Chem-Bet) with a loading of 100 mg per sample and a detector current of 127.0 mA. Catalyst samples for CO₂-TPD are *in-situ* reduced at 550 °C and then cooled down to 50 °C. Adsorption of CO₂ is conducted at 50 °C using a mixed gas of 2000 ppm CO₂ in Ar, followed by helium flushing at 50 °C for the removal of physically sorbed CO₂. The CO₂-TPD tests are conducted from ca. 50 °C to ca. 800 °C at 10 °C/min.

Catalytic Performance Tests

The Fischer-Tropsch to light olefins experiments are carried out in a fixed-bed tubular reactor. Firstly, catalysts (40–60 mesh in oxide status, 4 mL each) are loaded and reduced at 550 °C for 6 h by 50 % H₂/50 %Ar at atmospheric pressure. The weight is 4.2481 g, 3.4520 g, 3.6495 g, 2.3547 g, 4.9065 g and 4.0106 g for loaded catalysts F1, F2, F3, F_∞, REF-a and REF-b, respectively. The *in-situ* reduced catalysts are tested with a pre-mixed syngas (H₂/CO/Ar = 45/45/10, by mole%) at a total pressure of 2.0 MPa (i.e., approx-

imately 20 bar) and a space velocity (SV) of 1500 mL·g cat⁻¹·h⁻¹. Results are obtained after the reactions reach steady state. Tested reaction temperature are varied from low to high during catalytic tests. Argon is used as the internal standard component of syngas for GC analysis. The GC measurement method is similar to our previous reports.^[16–18,30,31] CO, CH₄ and CO₂ are analyzed by a GC with a TCD detector equipped with a TDX-01 packed column (6 m in length). Incondensable gaseous hydrocarbons are analyzed by a GC with a FID detector equipped with a PLOT Al₂O₃/S (50 m*0.53 mm 25 μm) capillary column. Water phase is analyzed by a GC with a TCD detector equipped with a GDX-102 (2 m*3 mm) packed column. Oxygenates in water phase are negligible and thus not counted in calculations. Condensed oil and wax (dissolved in cyclohexane) are analyzed by a GC with a FID detector equipped with an OV-101 (50 m*0.25 mm*0.5 μm) capillary column. Transport phenomenon may impact the product selectivity through limiting diffusion of reactants and products to and from catalytic sites.^[32,33] To minimize the effect of transport, the catalytic tests of catalysts are conducted under uniform loading volume (i.e., 4 mL, equal for each catalyst) as well as the same pellet size range of —40–60 mesh.

2. Results and Discussion

2.1. Structure Features. Freshly Reduced Catalysts vs. Used Catalysts

The catalysts for characterization are reduced at 550 °C for 6 h by 50% H₂/50% Ar. As shown in Table 1, support materials SP-0~SP-∞ have an increasing *x* value of 0~∞, where the value of *x* equals the molar ratio of nominal Al₂O₃ to nominal ZnAl₂O₄ in each of their nominal compositions. Supports SP-0 and SP-∞ were structurally analyzed for catalysts as well as their supports.

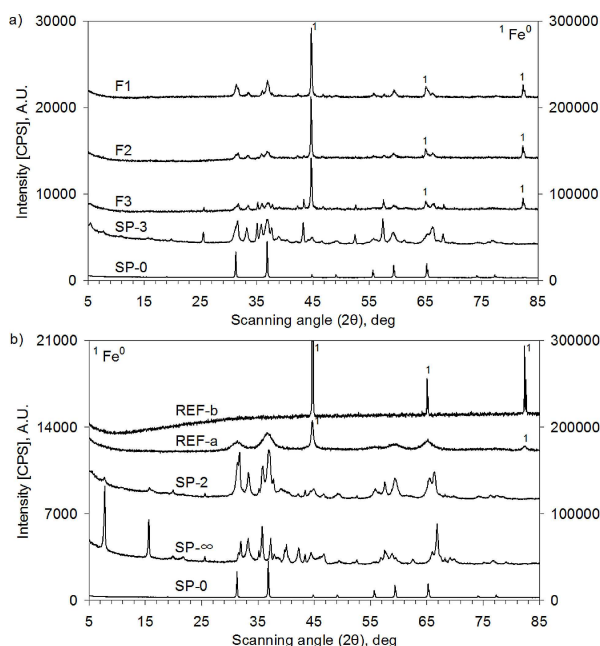


Figure 1. X-ray diffraction (XRD) patterns of supports and freshly reduced catalysts. Pattern “SP-0” follows the scale of the right ordinate axis, while others follow the scale of the left ordinate axis.

The XRD patterns of supports and freshly reduced catalysts are shown in Figure 1. The XRD patterns of used catalysts (i.e., discharged after catalytic tests) are shown in Figure S1. Detailed XRD patterns are shown in Supporting Information Figures S2~S6.

As shown in Figures 1&S2 and Table 1, support material SP-0 consists of ZnAl₂O₄ spinel phase (JCPDS 05-0669). The major alumina phase of support material SP-∞ is very close to β-Al₂O₃ (JCPDS 31-1263, 10-414, 51-769), while the minor alumina phase of support SP-∞ is α-Al₂O₃ (JCPDS 05-0712), which are both refractory materials. Beta-alumina, denoted as β-Al₂O₃ and β''-Al₂O₃, is one of two typical kinds of hexaaluminates (i.e., hexagonal aluminates including beta-alumina and magnetoplumbite) which are known to have a layer structure composed of conduction layers and spinel blocks that are stacked alternately.^[34,35] Cations such as Na⁺, K⁺, Ba²⁺ etc. may be accommodated in the conduction layer of beta-alumina through high temperature treatments.^[34,35] As listed in Table S1, the content of residual sodium (Na) in support SP-∞ is determined to be 2.49% by ICP-OES. The source of sodium in support SP-∞ is the residue of precipitant NaHCO₃. Ammonia salts (e.g., NH₄HCO₃) were not chosen as the precipitant because they may form some soluble complex ions with metal cations and cause the leaching of some metal atoms during filtering. NaHCO₃ is chosen as a precipitant that will not cause the leaching of metal atoms and can generate excellent mixing homogeneity.^[36] Unlike potassium (K), the promoting effect of sodium (Na) alone is fairly feeble, although sulfur plus sodium (i.e., a combination of S/Na) can generate synergetic effects on catalytic performance of iron catalysts.^[5–7]

With the help of structural analysis of SP-0 and SP-∞, the structure of other supports can be resolved. Table S1 shows that the content of sodium (1.19%) in support SP-1 is less than that in support SP-∞ (i.e., 2.49%). Figures S1&S3 and Table 1 show that support SP-1 is composed of ZnAl₂O₄ (major level), θ-Al₂O₃ (medium level, close to JCPDS 04-0877) and β-Al₂O₃ (few level). The description of major, medium, minor and low levels in this article depends on the XRD intensity of different phases. Freshly reduced catalyst F1 consists of Fe⁰ (major level, JCPDS 06-0696), ZnAl₂O₄ (medium level) and θ-Al₂O₃ (medium level). After catalytic tests, used catalyst F1 consists of ZnAl₂O₄ (major level) and θ-Al₂O₃ (medium level), while the status of iron species is X-ray amorphous.

Table S1 shows that the content of sodium (1.48%) in support SP-2 is less than that in support SP-∞ (i.e., 2.49%). Figure S4 and Table 1 shows that support SP-2 is composed of ZnAl₂O₄ (major level), θ-Al₂O₃ (major level), β-Al₂O₃ (medium level) and α-Al₂O₃ (minor level). Freshly reduced catalyst F2 consists of Fe⁰ (major level), ZnAl₂O₄ (minor level), θ-Al₂O₃ (minor level) and α-Al₂O₃ (low level). Although existed in support SP-2, crystalline β-Al₂O₃ phase becomes unobservable in the XRD pattern of freshly reduced catalyst F2. After catalytic tests, used catalyst F2 consists of ZnAl₂O₄ (major level), θ-Al₂O₃ (major level), α-Al₂O₃ (minor level) and β-Al₂O₃ (minor level), while the status of iron species is X-ray amorphous.

Table S1 shows that the content of sodium (1.18%) in support SP-3 is also less than that in support SP-∞ (i.e., 2.49%).

Figure S5 and Table 1 show that support SP-3 is composed of ZnAl_2O_4 (major level), $\alpha\text{-Al}_2\text{O}_3$ (major level), $\theta\text{-Al}_2\text{O}_3$ (major level) and $\beta\text{-Al}_2\text{O}_3$ (minor level). Freshly reduced catalyst F3 consists of Fe^0 (major level), ZnAl_2O_4 (medium level), $\theta\text{-Al}_2\text{O}_3$ (medium level) and $\alpha\text{-Al}_2\text{O}_3$ (medium level). Although existing in support SP-3, crystalline $\beta\text{-Al}_2\text{O}_3$ phase also becomes unobservable in the XRD pattern of freshly reduced catalyst F3. After catalytic tests, the $\beta\text{-Al}_2\text{O}_3$ phase becomes observable and thus used catalyst F3 consists of ZnAl_2O_4 (major level), $\theta\text{-Al}_2\text{O}_3$ (major level), $\alpha\text{-Al}_2\text{O}_3$ (major level) and $\beta\text{-Al}_2\text{O}_3$ (minor level), while the status of iron species is X-ray amorphous.

Figure S6 and Table 1 show that freshly reduced catalyst $\text{F}\infty$ consists of Fe^0 (major level), $\beta\text{-Al}_2\text{O}_3$ (major level) and $\alpha\text{-Al}_2\text{O}_3$ (minor level). After catalytic tests, used catalyst $\text{F}\infty$ consists of $\beta\text{-Al}_2\text{O}_3$ (major level) and $\alpha\text{-Al}_2\text{O}_3$ (minor level), while the status of iron species is X-ray amorphous.

The above data indicate that Al–O species (i.e., in the case of support SP- ∞) may tend to react with sodium at 1200 °C and generate the Na- $\beta\text{-Al}_2\text{O}_3$ phase (i.e., typically $\text{NaAl}_{11}\text{O}_{17}$), whereas superfluous (i.e., remaining) Al–O species may tend to form the $\alpha\text{-Al}_2\text{O}_3$ phase simultaneously. Similar phenomena happen for the Al–O species in the cases of supports SP-1, SP-2 and SP-3. Support SP-3 contains more superfluous Al–O species than supports SP-2 and SP-1, and, as a result, support SP-3 contains more $\alpha\text{-Al}_2\text{O}_3$ phase than supports SP-2 and SP-1. Moreover, zinc-doping may retard the transformation of Al–O species from the $\theta\text{-Al}_2\text{O}_3$ phase to the $\alpha\text{-Al}_2\text{O}_3$ phase, as shown in the crystalline phases of supports SP-1, SP-2, and SP-3. Above structure analyses indicate that the location of residue Na of catalysts $\text{F1}\sim\text{F}\infty$ is probably inside the 1200 °C annealed alumina phases (e.g., $\beta\text{-Al}_2\text{O}_3$ or its precursor, the $\theta\text{-Al}_2\text{O}_3$) that may lock sodium atoms as part of these phases, while the location of Zn of catalysts $\text{F1}\sim\text{F3}$ is inside the zinc aluminate phase.

Freshly reduced reference catalyst REF-a is composed of Fe^0 and ZnAl_2O_4 , while its alumina species remains X-ray amorphous. After catalytic tests, used reference catalyst REF-a consists of ZnAl_2O_4 phase only, while its iron species and alumina species both remain X-ray amorphous. Freshly reduced reference catalyst REF-b consists of a Fe^0 phase. After catalytic tests, used reference catalyst REF-b consists of the Fe_3C_2 phase (JCPDS No. 51-0997), which is consistent with the literature.^[37]

XPS is an effective tool to configure the state of iron on solid materials,^[38,39] and is thus employed to measure the state of iron of used catalyst F3 (Figures S7&S8). As shown in Figure S8a, curve fitting of this Fe2p spectrum shows that both Fe^{3+} and Fe^{2+} species are present on the surface of used catalyst F3. The BE (binding energy) value of the strongest Fe 2p_{3/2} band is 710.75 eV, which corresponds to the characteristic value of Fe_3O_4 .^[39] Other Fe2p bands at 712.75, 715.96, 719.37, 724.44 and 727.80 eV are respectively attributed to Fe^{3+} tet 2p_{3/2}, Fe^{2+} sat 2p_{3/2}, Fe^{3+} sat 2p_{3/2}, Fe^{3+} oct 2p_{1/2} and Fe^{2+} sat 2p_{1/2}, where the terms tet, oct and sat respectively denote tetrahedral, octahedral and satellite. The Fe 2p_{3/2} peak for iron carbide at 707.85 eV is very tiny and negligible. This result is consistent with the literature that the major iron state of syngas-treated iron catalysts is Fe_3O_4 , while the carburization degree of iron is

low (i.e., significantly lower than that of pure CO-treated iron catalysts) due to the oxidation of steam in FT conditions.^[40,41] As shown in Figure S8b, the BE value of the strongest C1s band is 284.67 eV, which corresponds to “graphitic” carbon which may also include adsorbed hydrocarbon products and surface precursors to hydrocarbon products.^[39] The C1s band at 285.17 eV is also attributed to the “graphitic” carbon. Other C1s bands at 285.65, 286.27 and 290.08 eV are attributed to carbonates.^[38,39] These carbon species are commonly present on the surface of syngas-treated iron catalysts.^[39]

2.2. Texture Features

As shown in Figure S9, the porous texture features of catalysts $\text{F1}\sim\text{F}\infty$ (thoroughly reduced at 550 °C) were determined by nitrogen sorption tests. The isotherm plots (Figure S9a) of catalysts $\text{F1}\sim\text{F}\infty$ are all of type IV as classified by IUPAC,^[42] suggesting that these catalysts are mesoporous adsorbents. The hysteresis loops of catalysts $\text{F1}\sim\text{F}\infty$ are all similar to type H3, suggesting the presence of mesoporosities formed with aggregates of plate-like particles giving rise to slit-shaped pores.^[42] The pore size distributions (Figure S9b) of catalysts F1 , F2 and F3 seem a little narrower than that of catalyst $\text{F}\infty$. The BET specific surface area values of catalysts F1 , F2 , F3 and $\text{F}\infty$ are 19.1, 19.6, 16.9 and 17.4 m²/g, respectively. The major pore sizes of catalysts F1 , F2 , F3 and $\text{F}\infty$ are 1.5, 1.8, 1.7 and 2.1 nm, respectively. The total pore volume values of catalysts F1 , F2 , F3 and $\text{F}\infty$ are 0.12, 0.12, 0.09 and 0.06 cm³/g for pores with diameter less than 138.2 nm at $P/P_0=0.985910$, respectively. The average pore diameters of catalysts F1 , F2 , F3 and $\text{F}\infty$ are 25.4, 25.4, 20.3 and 14.1 nm, respectively. Above data show that, as the content of Al_2O_3 in this type of catalyst increases, the average pore diameter and total pore volume may decrease. Above BET results of catalyst F3 is different from those of catalysts F1 and F2 , probably due to the circumstance that catalyst F3 contains more $\alpha\text{-Al}_2\text{O}_3$ phase than catalysts F1 and F2 .

2.3. Electron Microscopy Studies

FESEM was employed to observe the morphologies of freshly reduced catalysts. As shown in Figures 2a–c, catalyst F1 is composed of partially transparent flakes and particles, of which the flakes can be ascribed to alumina. These alumina flakes have a quadrangular or polygonal shape with an edge length of ca. 200–500 nm as well as a thickness of ca. 50 nm. The particles over these flakes are irregularly spherical and have a size of ca. 20–50 nm, probably due to Fe^0 and ZnAl_2O_4 crystals.

As shown in Figures 2d–f, similar to catalyst F1 , catalyst F2 is also composed of partially transparent flakes and particles. However, the alumina flakes of catalyst F2 seem less transparent than catalyst F1 , probably due to their differences in thickness. The alumina flakes of catalyst F2 have a quadrangular or polygonal shape with an edge length of ca. 200–1000 nm as well as a thickness of ca. 100 nm. The particles over these flakes

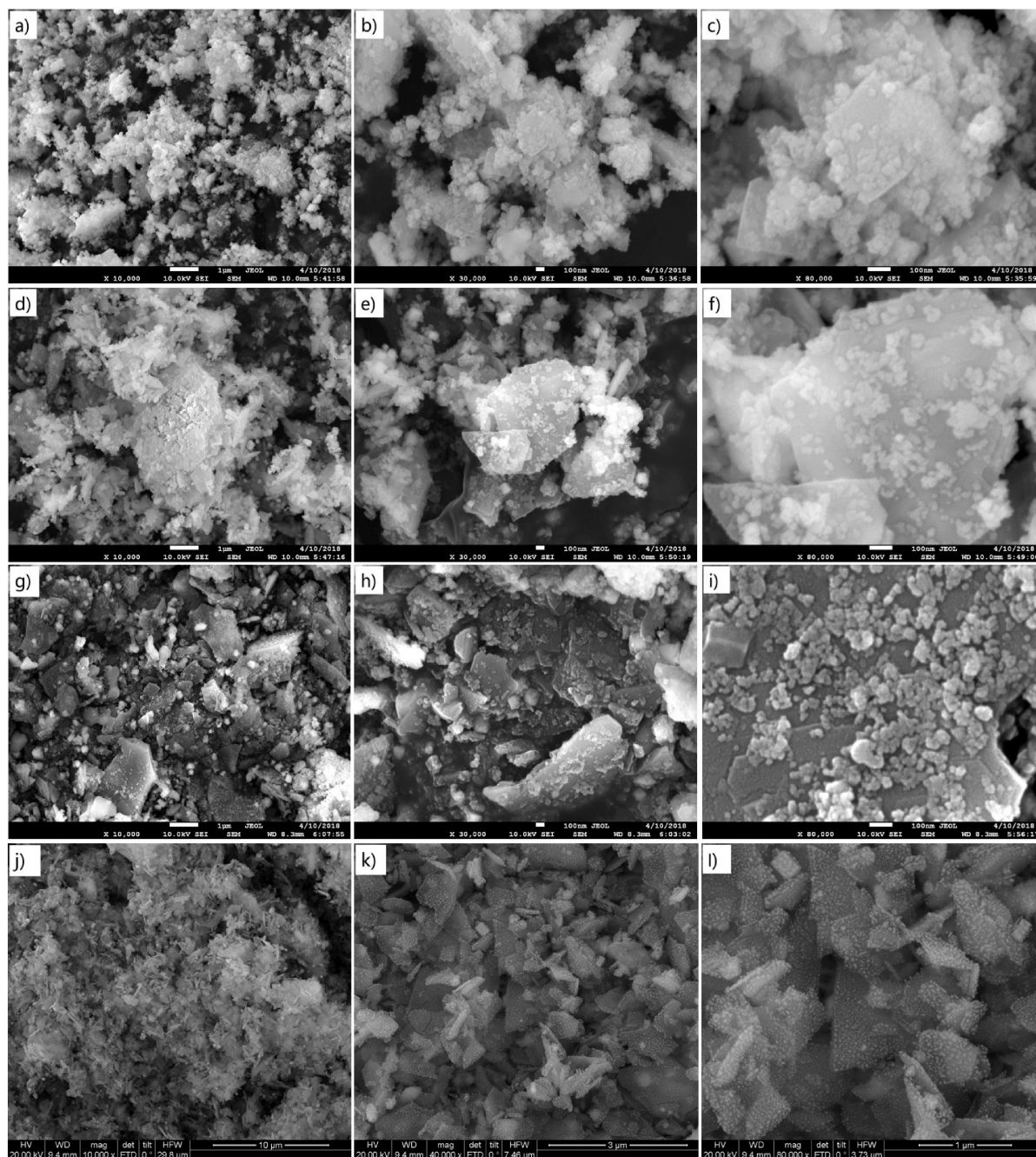


Figure 2. FESEM images of freshly reduced catalysts. a), b), c): catalyst F1. d), e), f): catalyst F2. g), h), i): catalyst F3. j), k), l): F_{∞} .

seem irregularly spherical and have a size of ca. 20–40 nm, and are also probably Fe^0 and ZnAl_2O_4 crystals.

As shown in Figures 2g–i, different from catalysts F1 and F2, catalyst F3 is composed of nontransparent plates and particles. The alumina plates of catalyst F3 have a polygonal or quadrangular shape with an edge length of ca. 300–2000 nm as well as a thickness of ca. 150–250 nm. The particles over these thick plates seem irregularly spherical and have a size of ca. 20–50 nm, also probably Fe^0 and ZnAl_2O_4 crystals.

As shown in Figures 2j–l, similar to catalysts F1 and F2, catalyst F_{∞} is also composed of partially transparent flakes and

particles, of which the flakes are mainly $\beta\text{-Al}_2\text{O}_3$ plus a little $\alpha\text{-Al}_2\text{O}_3$. The alumina flakes of catalyst F_{∞} have an irregular 2D shape with an edge length of ca. 100–1000 nm as well as a thickness of ca. 80–100 nm. The Fe^0 particles clearly deposited on alumina flakes have a size of ca. 15–30 nm.

The thickness of alumina flakes/plates seems related to the content of $\alpha\text{-Al}_2\text{O}_3$ phase. If considerable alumina has transformed to $\alpha\text{-Al}_2\text{O}_3$ phase (i.e., the case of F3), the morphology of alumina may become thick and nontransparent plates. If none or only small amounts of alumina have been transformed

to α - Al_2O_3 phase (i.e., the case of F1, F2 or F_∞), the morphology of alumina may remain as thin and partially transparent flakes.

The EDX elemental mapping results of freshly reduced catalyst F3 is shown in Supporting Information Figure S10. Iron particles are separately dispersed with a size range of 10–30 nm. Most iron particles have a size of 10–20 nm.

2.4. Surface Basicity

CO_2 -TPD is widely used to measure the surface basicity of catalysts.^[16,43] The CO_2 -TPD profiles of *in-situ* reduced Zn-containing catalysts are presented in Figure 3, Supporting Information Figure S11 and Table S2. The high temperature peak of catalyst F3 as well as high/low temperature peaks of catalysts REF-a and F_∞ look intense and significant, while other peaks in Figures 3 and S11 are weak and hump-like.

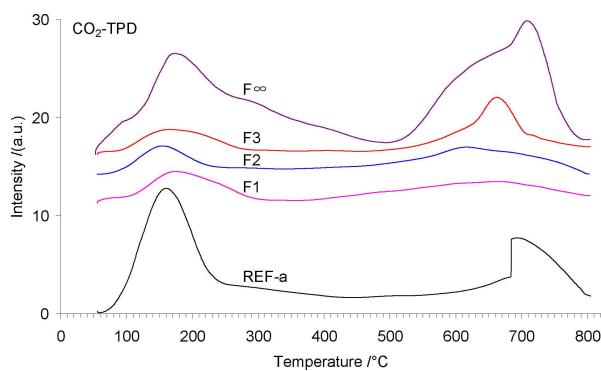


Figure 3. CO_2 temperature-programmed desorption (CO_2 -TPD) of freshly reduced Zn-containing catalysts.

Freshly reduced reference catalyst REF-a, carrying a $\text{ZnAl}_2\text{O}_4 \cdot 1\text{Al}_2\text{O}_3$ support calcined at 350°C , has two groups of CO_2 desorption peaks: one at ca. 160°C (i.e., the maximum value of its low temperature peak) corresponding to the desorption of weakly adsorbed CO_2 , while the other at ca. 693°C (i.e., the maximum value of its high temperature peak) is ascribed to the desorption of strongly adsorbed CO_2 , indicating that the surface of catalyst REF-a has both weak basic sites for loose CO_2 /catalyst binding and strong basic sites for tight CO_2 /catalyst binding. The area of high temperature peak of catalyst REF-a is significantly smaller than the low temperature peak, whilst the intensity of high temperature peak of catalyst REF-a is significantly lower than the low temperature peak. The total integral area for catalyst REF-a is 16970 (A.U.). The CO_2 -TPD curve of reference catalyst REF-a^[31] is permitted by copyright of Springer Nature.

Catalyst F1, with a $\text{ZnAl}_2\text{O}_4 \cdot 1\text{Al}_2\text{O}_3$ support calcined at 1200°C , has two groups of CO_2 desorption peaks: one at ca. 174°C corresponding to the desorption of weakly adsorbed CO_2 , while the other at ca. 661°C is ascribed to the desorption of strongly adsorbed CO_2 . The area of high temperature peak of catalyst F1 is significantly larger than the low temperature peak,

while the intensity of high temperature peak of catalyst F1 is significantly lower than the low temperature peak. The total integral area for catalyst F1 is 10518 (A.U.). By comparing the CO_2 -TPD data of catalysts REF-a and F1 which have the same nominal composition, one may find that the high temperature treatment at 1200°C may effectively lower down the intensities of both the low temperature peak and the high temperature peak, but meanwhile will not change the sequence of peak intensities (i.e., a more intense peak will still be more intense no matter the passivation treatment). The total integral area of catalyst F1 is significantly less than that of catalyst REF-a, indicating that the high temperature passivation at 1200°C may effectively decrease the amount of total basic sites.

Catalyst F2, with a $\text{ZnAl}_2\text{O}_4 \cdot 2\text{Al}_2\text{O}_3$ support calcined at 1200°C , has two groups of CO_2 desorption peaks: one at ca. 154°C corresponding to the desorption of weakly adsorbed CO_2 , while the other at ca. 616°C is ascribed to the desorption of strongly adsorbed CO_2 . The area of high temperature peak of catalyst F2 is significantly larger than the low temperature peak, while the intensity of high temperature peak of catalyst F2 is similar to the low temperature peak. The total integral area for catalyst F2 is 9809 (A.U.).

Catalyst F3, with a $\text{ZnAl}_2\text{O}_4 \cdot 3\text{Al}_2\text{O}_3$ support calcined at 1200°C , has two groups of CO_2 desorption peaks: one at ca. 165°C corresponding to the desorption of weakly adsorbed CO_2 , while the other at ca. 662°C is ascribed to the desorption of strongly adsorbed CO_2 . The area of high temperature peak of catalyst F3 is significantly larger than the low temperature peak, whilst the intensity of high temperature peak of catalyst F3 is significantly higher than the low temperature peak. The reason why the high temperature peak of catalyst F3 is significantly higher than the low temperature peak still remains unclear. The total integral area for catalyst F3 is 10683 (A.U.).

Catalyst F_∞ , with an Al_2O_3 support calcined at 1200°C , has two groups of CO_2 desorption peaks: one at ca. 173°C corresponding to the desorption of weakly adsorbed CO_2 , while the other at ca. 708°C is ascribed to the desorption of strongly adsorbed CO_2 . The area of high temperature peak of catalyst F_∞ is slightly smaller than the low temperature peak, while the intensity of high temperature peak of catalyst F_∞ is a little higher than the low temperature peak. The total integral area for catalyst F_∞ is 21466 (A.U.).

2.5. Catalytic Performance Tests. Catalytic Activity

As shown in Figure 4, the CO conversions of all catalysts increase with elevated reaction temperature. A comparison of the deduced temperatures at which the CO conversions may reach 50% shows that the catalytic activities of catalysts for CO conversion have a declining sequence of $\text{REF-b} > F_\infty \approx F2 \approx \text{REF-a} > F1 > F3$. Catalyst REF-b, the unsupported 100% iron, owns more iron atoms and behaves more active than other catalysts that contain only 15% supported iron.

By comparing the CO conversion data of catalysts REF-a and F1 which have the same nominal composition, one may find that the high temperature treatment at 1200°C may lower

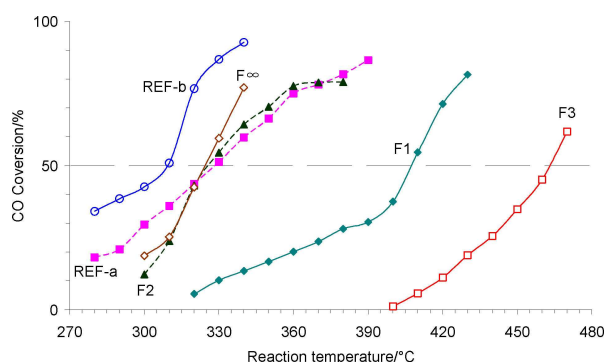


Figure 4. Comparison of CO conversions for reduced catalysts at different reaction temperatures.

down the catalytic activity to some extent. The FTO reaction rate depends on the dissociative CO activation.^[44] When CO molecules adsorb onto the surface of iron, the CO molecules tend to draw electrons from the iron.^[45] As determined by the A_{total} values of CO_2 -TPD results in Supporting Information Table S2, the partial passivation of support phase at 1200 °C makes catalyst F1 own significantly less amount of total basic sites than catalyst REF-a. Thus catalyst F1 may donate less electrons for dissociative CO activation and, as a result, produce a lower CO conversion than catalyst REF-a.

As shown in Supporting Information Figure S12 and Tables S3~S8, at the reaction temperature of 320 °C, the FTY (iron time yield, $10^{-5} \text{ mol}_{\text{CO}}/\text{g}_{\text{Fe}}\text{s}$) values can be compared and show a decreasing sequence as follows: REF-a(3.9) > F_{∞} (3.8) > F2(3.7) > REF-b(1.0) > F1(0.5) > F3(0). Although the CO conversion of catalyst REF-b is the highest one at 320 °C, it is accomplished by 100%Fe while other catalysts own only 15% Fe. By comparing the CO conversion data and FTY data of catalysts F1, F2, F3 and F_{∞} , there is no linear relationship between catalytic activity of unit iron and the x value.

Selectivity to CO_2

In Fischer-Tropsch reaction, the theoretical selectivity to CO_2 ought to equal to or below 50%. Most results in Supporting Information Tables S3~S8 follow this theoretically deduced data. However, several data on selectivity to CO_2 are a little higher than 50% (e.g., catalyst F1 at 320~340 °C, catalyst F2 at 300~310 °C). According to the theory of M. Ojeda and E. Iglesia et al.,^[46] these data indicate that, other than H-assisted CO dissociation that removes O-atoms as H_2O , these two catalysts (i.e., catalysts F1 and F2) may either behave rather active in direct CO dissociation that removes O-atoms as CO_2 , or behave rather active in water-gas shift reaction.

Hydrocarbon Product Distribution (C_{5+})

Figures 5~9 and Supporting Information Figure S13 show the comparisons of catalysts' hydrocarbon product distributions

(i.e., also called hydrocarbon selectivity or CH selectivity) based on C molar content, representing the selectivities of C_{5+} , C_1 , C_2 - C_4 and so forth in hydrocarbons, respectively. Both types of comparisons (vs. temperature, vs. CO conversion) are provided in Figures 5~9 and Supporting Information Figure S13. Catalysts' detailed data of catalytic performance are summarized in Supporting Information Tables S3~S8.

Usually, practical CO hydrogenation processes are not operated at very high CO conversions because high conversion causes high H_2O partial pressure, which accelerates catalyst decay due to the formation of inactive metal-support compounds in harsh steam environments.^[47] A moderate CO conversion is therefore a practical choice.^[47] The CO conversion of 50% is referred to in this article as a moderate operational condition for general comparisons. The deduced selectivity values of different catalysts at CO conversion of ca. 50% are employed for the observation of selectivities of individual catalysts. Thus the comparison of product selectivities at the same/similar CO conversion is an alternative route and is acceptable such as in our recent publications.^[16-18,30]

As shown in Figure 5, the C_{5+} hydrocarbon distribution values of all catalysts except catalyst F_{∞} generally decrease with elevated reaction temperature or CO conversion, because raising the reaction temperature may accelerate the desorption rate of intermediate products and thus suppress chain propagation reactions. As shown in Figure 5d, at the moderate CO conversion of 50%, the C_{5+} hydrocarbon distribution values of catalysts F1, REF-a, and REF-b have a decreasing sequence of REF-b > REF-a > F1, suggesting that unsupported 100% iron (i.e., catalyst REF-b) is favorable for carbon chain propagation, while $\text{ZnAl}_2\text{O}_4 \cdot 1\text{Al}_2\text{O}_3$ supported iron catalysts (i.e., REF-a and F1) may suppress chain propagation reactions to some extent. These data also verify that high temperature treatment at 1200 °C (i.e., catalyst F1) may suppress the C-C coupling more effectively than the low temperature treatment at 350 °C (i.e., catalyst REF-a).

As shown in Figure 5b, at a moderate CO conversion of 50%, the C_{5+} hydrocarbon distribution values of catalysts F1~ F_{∞} have a decreasing sequence of F_{∞} > F2 > F1 $\approx$ F3, indicating that the capability of suppressing chain propagation is not linearly related with the x value. These data also indicate that catalysts F1 and F3 may suppress C-C coupling more effectively than catalysts F2 and F_{∞} .

Hydrocarbon Product Distribution (C_1 - C_4)

The C_1 - C_4 hydrocarbon distribution equals the difference between 100% and C_{5+} hydrocarbon distribution. As shown in Supporting Information Figure S13, a high C_1 - C_4 hydrocarbon distribution, as is observed in the data of catalysts F3 and F1, is favored in FTO studies due to the inclusion of C_2 - C_4 hydrocarbon distribution. If the C_1 - C_4 hydrocarbon distribution is low, then the C_2 - C_4 hydrocarbon distribution will not be high.

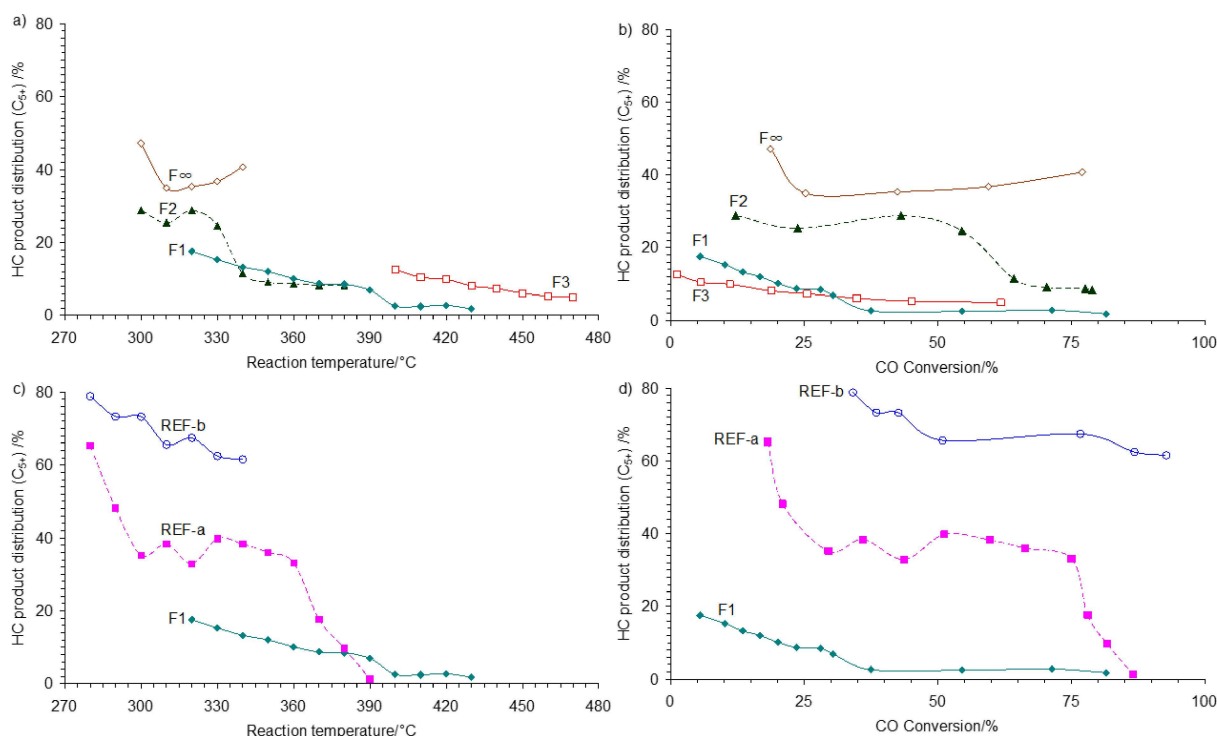


Figure 5. C_{5+} hydrocarbon distributions (i.e., C_{5+} CH selectivity) of reduced catalysts at different reaction temperatures (a, c) and different CO conversions (b, d). Effect of x value is compared in (a, b), while effect of calcinations is compared in (c, d).

Hydrocarbon Product Distribution (CH_4)

As shown in Figure 6, the C_1 (i.e., CH_4) hydrocarbon distribution values of all catalysts generally increase with elevated reaction temperature or CO conversion. As shown in Figure 6d, at the moderate CO conversion of 50%, the C_1 (i.e., CH_4) hydrocarbon distribution values of catalysts F1, REF-a and REF-b have an increasing sequence of REF-b < REF-a < F1. As shown in Figure 6b, at the moderate CO conversion of 50%, the C_1 (i.e., CH_4) hydrocarbon distribution values of catalysts F1~F ∞ have an increasing sequence of F ∞ < F2 < F3 < F1. The above trends of C_1 (i.e., CH_4) hydrocarbon distribution in Figure 6 are quite similar to those of C_1 - C_4 hydrocarbon distributions in Supporting Information Figure S13, indicating that the amount of methane has positive correlation with the catalyst's capability of suppressing chain propagation. When the C-C coupling is suppressed, the C_{5+} hydrocarbon distribution will drop. When the long chain hydrocarbon (i.e., C_{5+}) distribution drops, the short chain hydrocarbon (i.e., C_1 and C_2 - C_4) distribution(s) ought to rise. Furthermore, as a fully hydrogenated product, the amount of CH_4 may be foreseen to be also affected by the catalyst's hydrogenation capability to some extent.

Hydrocarbon Product Distribution (C_2 - C_4)

As shown in Figure 7, the C_2 - C_4 hydrocarbon distribution values of catalysts REF-a and REF-b generally increase with elevated reaction temperature or CO conversion, while those of catalysts

F1~F3 remain either almost flat (i.e., catalyst F3) or fluctuate at narrow ranges (i.e., catalysts F1 and F2). As shown in Figure 7d, at the moderate CO conversion of 50%, the C_2 - C_4 hydrocarbon distribution values of catalysts F1, REF-a and REF-b have an increasing sequence of REF-b < REF-a < F1. As shown in Figure 7b, at the moderate CO conversion of 50%, the C_2 - C_4 hydrocarbon distribution values of catalysts F1~F ∞ have an increasing sequence of F ∞ < F2 < F1 < F3. The above sequences of C_2 - C_4 hydrocarbon distribution in Figure 7 are basically similar to those of the C_1 - C_4 hydrocarbon distribution in Supporting Information Figure S13 as well as those of the C_1 (i.e., CH_4) hydrocarbon distribution in Figure 6, indicating that the amount of C_2 - C_4 also has a positive correlation with the catalyst's capability of suppressing chain propagation. At CO conversion of 35%, catalyst F3 exhibits a C_2 - C_4 hydrocarbon distribution value of 67%, which significantly exceeds the ASF (Anderson-Schulz-Flory) C_2 - C_4 limit value of 58%.

C_2 - C_4 Olefin/paraffin Ratio

As shown in Figure 8, the C_2 - C_4 olefin/paraffin ratios of all catalysts generally decrease with elevated reaction temperature or CO conversion. As shown in Figure 8d, at the moderate CO conversion of 50%, the C_2 - C_4 olefin/paraffin ratios of catalysts F1, REF-a and REF-b have a decreasing sequence of REF-b > F1 $\approx$ REF-a. As shown in Figure 8b, at the moderate CO conversion of 50%, the C_2 - C_4 olefin/paraffin ratios of catalysts F1~F ∞ have a decreasing sequence of F3 > F ∞ > F2 > F1.

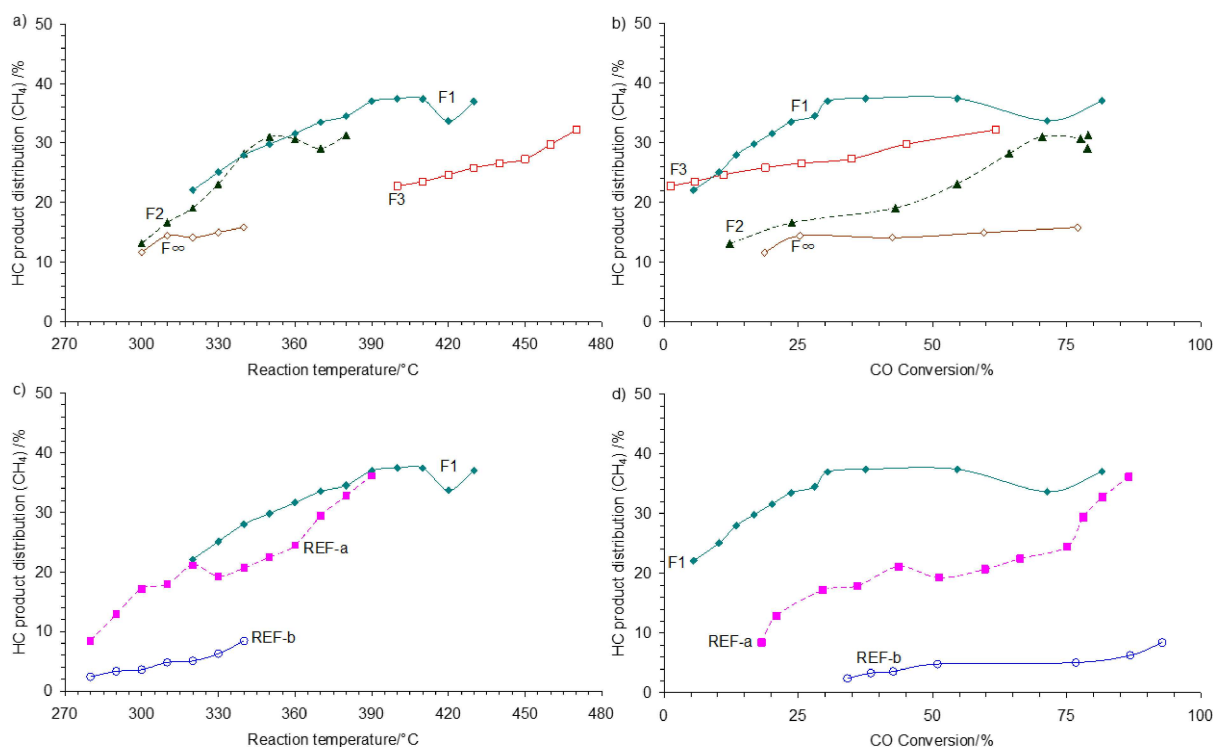


Figure 6. CH₄ hydrocarbon distributions (i.e., CH₄ CH selectivity) of reduced catalysts at different reaction temperatures (a, c) and different CO conversions (b, d). Effect of x value is compared in (a, b), while effect of calcinations is compared in (c, d).

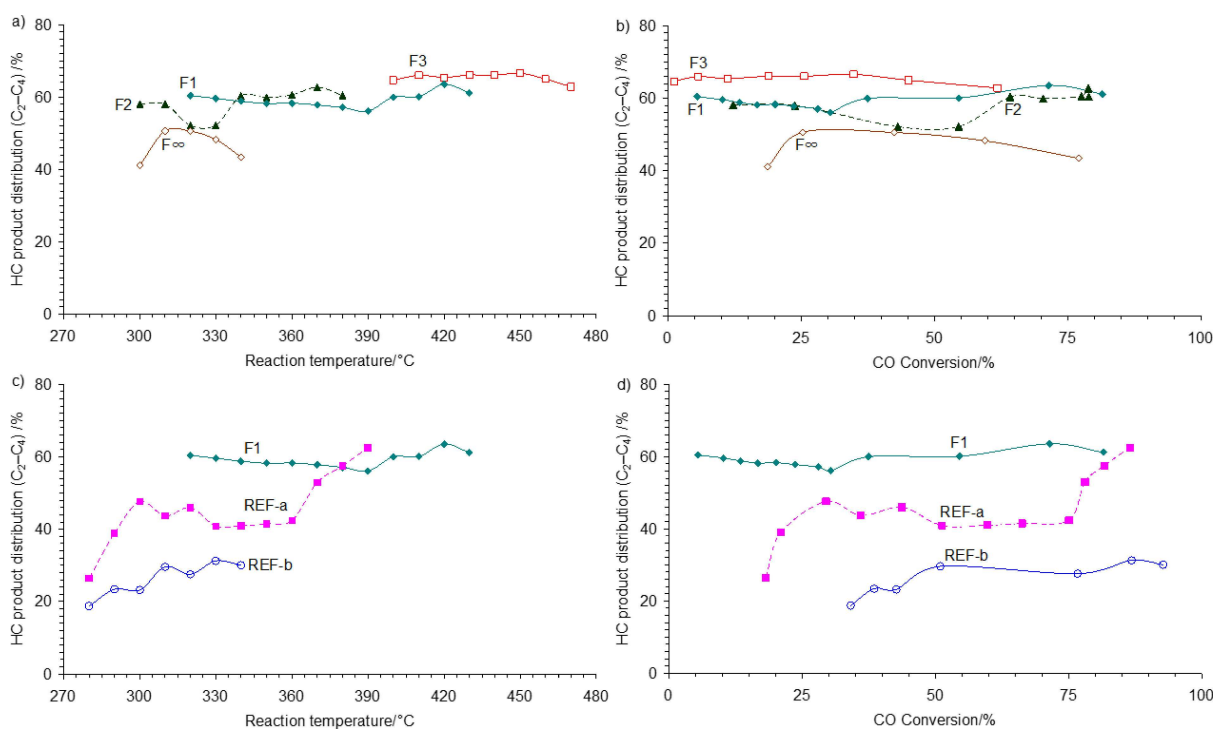


Figure 7. C₂-C₄ hydrocarbon distributions (i.e., C₂-C₄ CH selectivity) of reduced catalysts at different reaction temperatures (a, c) and different CO conversions (b, d). Effect of x value is compared in (a, b), while effect of calcinations is compared in (c, d).

Unsupported 100% iron (i.e., catalyst REF-b) retains a very high C₂-C₄ olefin/paraffin ratio above 7.0 even at a high CO

conversion of ca. 87%. The C₂-C₄ olefin/paraffin ratio of catalyst F1 is higher than that of REF-a at low CO conversions. Catalyst

F3 is the only supported catalyst that reaches a high C_2 - C_4 olefin/paraffin ratio above 7.0 at a CO conversion below ca. 19%. At a CO conversion of ca. 45%, catalyst F3 may still reach a high C_2 - C_4 olefin/paraffin ratio above 6.6. These data suggest that catalyst F3 has significantly better capability of suppressing hydrogenation than catalysts F1, F2, F_∞ and REF-a.

Hydrocarbon Product Distribution (C_2 - C_4)

The FTO process favors higher C_2 - C_4 (i.e., the C_2 - C_4 olefins) hydrocarbon distribution values.^[5] As shown in Figure 9d, at the moderate CO conversion of 50%, the C_2 - C_4 hydrocarbon distribution values of catalysts F1, REF-a and REF-b have a decreasing sequence of $F1 > REF-a \approx REF-b$. At the moderate CO conversion below 50%, this sequence becomes $F1 > REF-a > REF-b$. Although unsupported 100% iron (i.e., catalyst REF-b) has the highest C_2 - C_4 olefin/paraffin ratios, its C_2 - C_4 hydrocarbon distribution values are low and thus its C_2 - C_4 hydrocarbon distribution values are also low. By comparing the data of catalysts REF-a and F1, one may find that the high temperature treatment at 1200 °C may effectively enhance the C_2 - C_4 hydrocarbon distribution values.

Catalysts F_∞ , F1 and F2 reach their maximum C_2 - C_4 hydrocarbon distribution values of 40%, 47% and 49% at their CO conversions of 25%, 5% and 12%, respectively. Catalyst F3, which has a nominal composition of 15%Fe/85% ($ZnAl_2O_4 \cdot 3Al_2O_3$), exhibits a surprising C_2 - C_4 hydrocarbon distribution value of 58% at a total pressure of ca. 20 bar and a CO conversion of 35%. Generally, 100%Fe, Fe/ γ - Al_2O_3 , and Fe/

carbon black belong to conventional iron catalysts and usually have a maximum C_2 - C_4 hydrocarbon distribution below 30%^[48] at similar reaction conditions. The Fe/ β - Al_2O_3 herein (i.e., catalyst F_∞) reaches a maximum C_2 - C_4 hydrocarbon distribution of 40%. Without the help of any water-soluble efficient promoters (such as K or combination of S/Na), the C_2 - C_4 hydrocarbon distribution of 58% by catalyst F3 demonstrates that the 1200 °C passivated $ZnAl_2O_4 \cdot 3Al_2O_3$ support (i.e., SP-3) is probably the most selective support material for iron-based FTO catalysts by far, achieving a huge increase in C_2 - C_4 hydrocarbon distribution by almost 100% in comparison with regular supports (e.g., carbon, γ - Al_2O_3 , etc.). Through the high temperature passivation of support phase at 1200 °C, each formed component of a catalyst is expected to minimize the disturbances among them and thus exhibit their individual, unique, and appropriate functions.

Moreover, at the CO conversions below 45% as listed in Supporting Information Table S3, the olefin/paraffin (O/P) ratio of C_2 hydrocarbons (i.e., ethylene C_2H_4 and ethane C_2H_6) for catalyst F3 keeps in a range of 4.4~5.0, which almost reaches that for unsupported 100% Fe (i.e., catalyst REF-b) and is much higher than other supported iron catalysts (e.g., F1, F2, F_∞ , REF-a). In the case of catalyst F3, at a CO conversion of 35%, the contents of ethylene in C_2 (i.e., the C_2 olefin fraction), propylene in C_3 (i.e., the C_3 olefin fraction), and butylene in C_4 hydrocarbons (i.e., the C_4 olefin fraction) achieve 82%, 92%, and 91%, respectively. At a total pressure of 2.0 MPa (i.e., ca. 20 bar) where most FTO catalysts have a low C_2 O/P ratio of ca. 0.5~2.0,^[5,9] such a high C_2 O/P ratio of 4.4~5.0 is impressive and

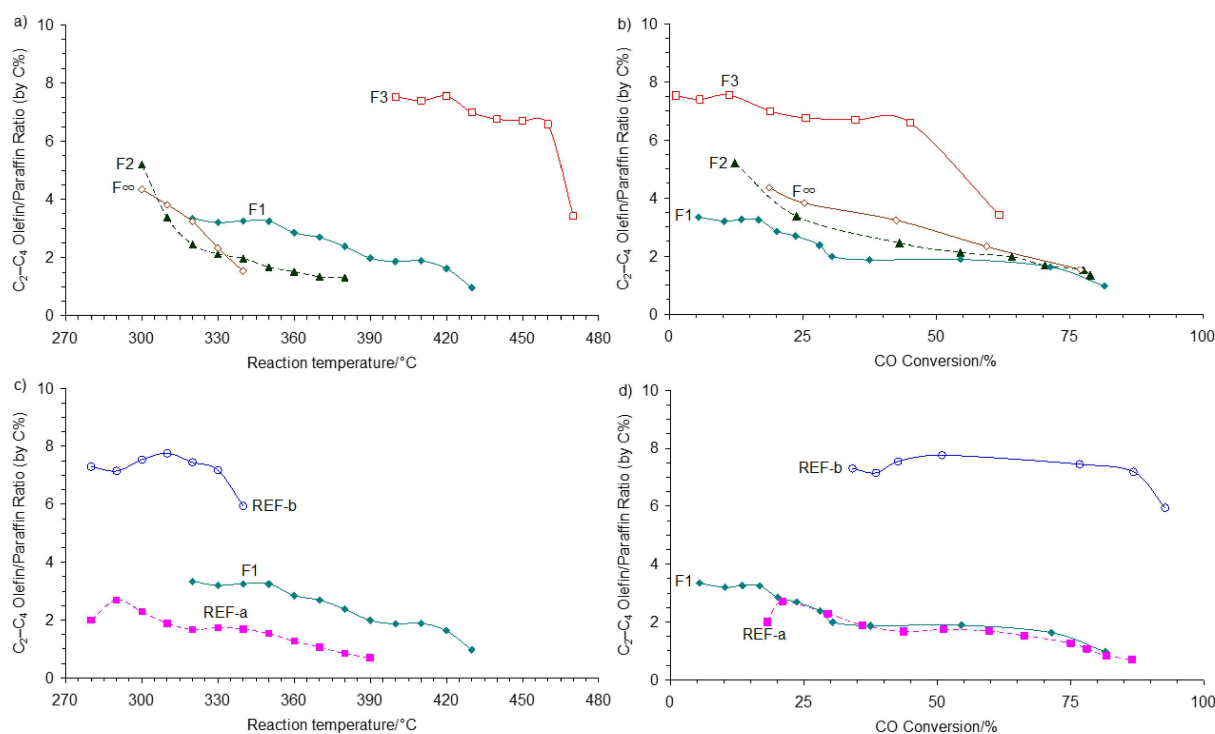


Figure 8. C_2 - C_4 olefin/paraffin ratios of reduced catalysts at different reaction temperatures (a, c) and different CO conversions (b, d). Effect of x value is compared in (a, b), while effect of calcinations is compared in (c, d).

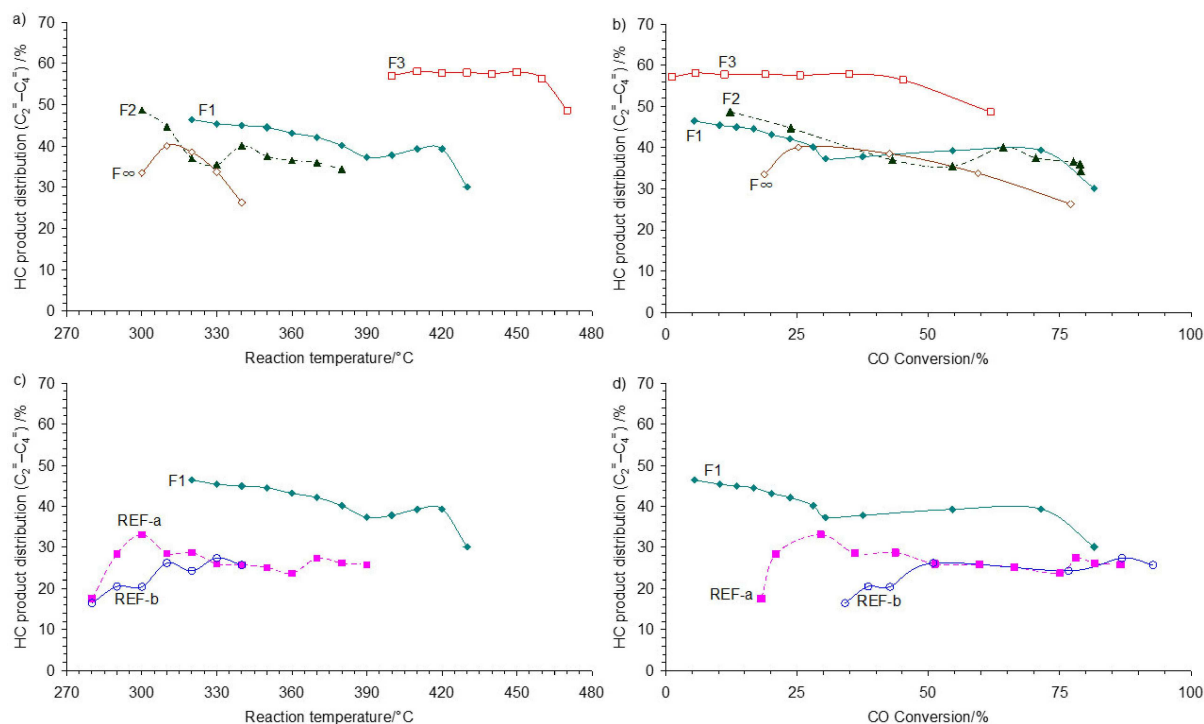


Figure 9. C₂-C₄ hydrocarbon distributions (i. e., C₂-C₄ = CH selectivity) of reduced catalysts at different reaction temperatures (a, c) and different CO conversions (b, d). Effect of *x* value is compared in (a, b), while effect of calcinations is compared in (c, d).

difficult to achieve for supported iron FTO catalysts due to the existence of secondary reactions such as further hydrogenation.

2.6. Further Discussions

As shown in Figures 8b, d and Supporting Information Table S2, at the moderate CO conversion of 50%, the C₂-C₄ olefin/paraffin ratios of all tested catalysts are found to be positively correlated with the values of $I_{\text{high}}/I_{\text{low}}$ which respectively denote the integral intensities of high temperature peaks and low temperature peaks of CO₂-TPD plots. For the comparison of catalysts with the same potassium (i. e., a water-soluble efficient promoter) content, the $I_{\text{high}}/I_{\text{low}}$ value of a catalyst may thus become a promising candidate as a probe for the capability of suppressing hydrogenation.

As shown in Figure 5b, compared with catalyst F_∞ which is ZnAl₂O₄-free, the existence of ZnAl₂O₄ phase in catalysts F1~F3 can effectively lower down the C₅₊ hydrocarbon distribution value, indicating that the ZnAl₂O₄ component in catalysts may function to suppress carbon chain propagation.

As shown in Figure 8b, the C₂-C₄ olefin/paraffin ratios of catalysts F1~F3 have a decreasing sequence of F3 > F2 > F1, which exhibit a positive correlation with the *x* value. Herein the passivated alumina phase may act as a solid base to some extent and function to suppress the hydrogenation ability. For the case of catalyst F_∞, it contains the highest amount of passivated alumina phase and thus behaves like a catalyst with

superfluous amount of base, by which an ordinary level C₂-C₄ olefin/paraffin ratio is caused.

M. Luo and B. H. Davis found that potassium (a strong base) promoter may effectively enhance the C₂ olefin/paraffin ratio and C₂ olefin fraction.^[49] As listed in Supporting Information Table S3, the high C₂ olefin/paraffin (O/P) ratio of catalyst F3 also demonstrates that the 1200 °C passivated alumina phase may act as a solid base which is similar to potassium to some extent (i. e., still not completely like K yet considering their different CO conversion trends), however, a good thing is that, unlike potassium, this 1200 °C passivated alumina phase is insoluble in water/steam conditions. As for the case of catalyst F_∞, it contains the highest amount of passivated alumina phase and thus behaves like a catalyst with superfluous amount of base, by which an ordinary level C₂ olefin/paraffin ratio is caused.

As mentioned above, as a CO molecule adsorbs onto an iron surface, it will tend to draw electrons from the iron. As a H₂ molecule adsorbs (at not very low coverages) onto an iron surface, it will tend to donate electrons to the iron.^[45] A base (e. g., K₂O) is electron donors that can donate electrons to neighboring iron atoms, strengthen the Fe-C bond, and decrease the hydrogenation ability.^[45] As shown in Figures 5b, d and Supporting Information Table S2, both catalyst F_∞ and catalyst REF-a have very high A_{total} values, representing that their total basic sites are significantly more than other catalysts. Catalyst F_∞ exhibits higher C₅₊ hydrocarbon distribution values than catalysts F1~F3, indicating that the basic sites may facilitate the covering of carbon species onto the catalyst

surface, reach necessary concentration of carbon species, and then grow for carbon chain propagation.

Catalyst REF-a and catalyst F1 have the same nominal composition. As shown in Figure 5b and Supporting Information Table S2, the high temperature passivation at 1200 °C (i.e., for catalyst F1) may effectively lower down the A_{total} value and total basic sites, and thus may suppress the carbon chain propagation and lower down the C_{5+} hydrocarbon distribution. The $I_{\text{high}}/I_{\text{low}}$ value (i.e., 0.7) of catalyst F1 is only slightly higher than that of catalyst REF-a, indicating that the major function of high temperature passivation at 1200 °C is not on suppression of hydrogenation ability but on suppression of carbon chain propagation. In the future, we will conduct studies in sequence on more detailed calcination temperature points, effects of efficient promoters like potassium, as well as a final life test for the application of catalysts.

3. Conclusions

A high temperature of 1200 °C has been employed to prepare partially passivated supports with nominal compositions of $\text{ZnAl}_2\text{O}_4 \cdot x\text{Al}_2\text{O}_3$ ($x=1, 2, 3$) for simplified iron/support CO hydrogenation catalysts. Zinc-doping may retard the transformation of Al–O species from the $\theta\text{-Al}_2\text{O}_3$ phase to the $\alpha\text{-Al}_2\text{O}_3$ phase during calcination at 1200 °C. The thickness of high temperature treated alumina flakes/plates is related to the content of the $\alpha\text{-Al}_2\text{O}_3$ phase. The crystalline oxide phases in freshly reduced catalysts remain stable after catalytic tests. Compared with a low calcination temperature of 350 °C, the high calcination temperature of 1200 °C for the Zn–O–Al supports can make catalysts achieve higher $C_2\text{--}C_4$ and $C_2^=\text{--}C_4^=$ hydrocarbon distribution values. The x value in nominal compositions can significantly affect the capability for suppression of hydrogenation.

The functions of zinc aluminate phase, alumina phase as well as the 1200 °C passivation treatment have been discussed with the following results. The $C_2\text{--}C_4$ olefin/paraffin ratios of all tested catalysts are found to be positively correlated with the values of $I_{\text{high}}/I_{\text{low}}$, which respectively denote the integral intensities of high temperature peaks and low temperature peaks of $\text{CO}_2\text{-TPD}$ plots. The ZnAl_2O_4 component in catalysts may function to suppress carbon chain propagation. The passivated alumina phase may act as a solid base to some extent and function to suppress the hydrogenation ability, especially in the effective enhancement of C_2 olefin/paraffin (O/P) ratio at an appropriate alumina usage. The major function of high temperature passivation at 1200 °C is not on suppression of hydrogenation ability but on suppression of carbon chain propagation.

Without the assistance of any water-soluble efficient promoters (such as K or combination of S/Na), at a total pressure of 20 bar and a CO conversion of 35%, catalyst F3 achieves a surprising $C_2^=\text{--}C_4^=$ hydrocarbon distribution of 58%, which is almost 100% higher than iron catalysts over regular supports (e.g., Al_2O_3 , carbon, etc.) and demonstrates the 1200 °C passivated $\text{ZnAl}_2\text{O}_4 \cdot 3\text{Al}_2\text{O}_3$ (i.e., SP-3) as probably the

most selective support material for iron-based FTO catalysts. The crystalline phases of this highly selective support include ZnAl_2O_4 , $\theta\text{-Al}_2\text{O}_3$, $\alpha\text{-Al}_2\text{O}_3$ plus a little $\beta\text{-Al}_2\text{O}_3$. At a CO conversion of 35%, this highly selective simplified catalyst (i.e., F3) can generate an ethylene content of 82% in C_2 , a propylene content of 92% in C_3 , and a butylene content of 91% in C_4 hydrocarbons. This research has led to development of chemically stable, structurally stable solid base-type supports and relevant iron-based catalysts for selective hydrogenation of carbon monoxide to light olefins.

Acknowledgments

We thank the National Natural Science Foundation of China (NSFC, No. 21571161, for Dr. Y. Xing) and Research Funding for Ph.D. Faculty of Zhengzhou University of Light Industry (No. 2016BSJJ034/CLY20170069/LZX2016, for Dr. Z. Liu) for financial support.

Conflict of Interest

The authors declare no conflict of interest.

Keywords: CO hydrogenation · Fischer Tropsch · light olefins · spinel · iron catalyst

- [1] J.-S. Bae, S. Y. Hong, J. C. Park, G. B. Rhim, M. H. Youn, H. Jeong, S. W. Kang, J.-I. Yang, H. Jung, D. H. Chun, *Appl. Catal. B* **2019**, *244*, 576–582.
- [2] H. G. Salazar-Contreras, A. Martinez-Hernandez, A. A. Boix, G. A. Fuentes, E. Torres-Garcia, *Appl. Catal. B* **2019**, *244*, 414–426.
- [3] Q.-X. Luo, L.-P. Guo, S.-Y. Yao, J. Bao, Z.-T. Liu, Z.-W. Liu, *J. Catal.* **2019**, *369*, 143–156.
- [4] J. M. Cho, S. Kasipandi, Y. M. Park, J. W. Bae, *Fuel*, **2019**, *239*, 547–558.
- [5] H. M. T. Galvis, K. P. de Jong, *ACS Catal.* **2013**, *3*, 2130–2149.
- [6] H. M. T. Galvis, J. H. Bitter, M. Ruitenbeek, K. P. de Jong, *Science* **2012**, *335*, 835–838.
- [7] H. M. T. Galvis, A. C. J. Koeken, J. H. Bitter, T. Davidian, M. Ruitenbeek, A. I. Dugulan, K. P. de Jong, *J. Catal.* **2013**, *303*, 22–30.
- [8] F. Jiao, J. Li, X. Pan, J. Xiao, H. Li, H. Ma, M. Wei, Y. Pan, Z. Zhou, M. Li, S. Miao, J. Li, Y. Zhu, D. Xiao, T. He, J. Yang, F. Qi, Q. Fu, X. Bao, *Science* **2016**, *351*, 1065–1068.
- [9] L. Zhong, F. Yu, Y. An, Y. Zhao, Y. Sun, Z. Li, T. Lin, Y. Lin, X. Qi, Y. Dai, L. Gu, J. Hu, S. Jin, Q. Shen, H. Wang, *Nature* **2016**, *538*, 84–87.
- [10] H. T. Luk, C. Mondelli, J. Perez-Ramirez, D. C. Ferre, J. A. Stewart, *Chem. Soc. Rev.* **2017**, *46*, 1358–1426.
- [11] G. Collinge, N. Kruse, J.-S. McEwen, *J. Phys. Chem. C* **2017**, *121*, 2181–2191.
- [12] M. Gupta, M. L. Smith, J. J. Spivey, *ACS Catal.* **2011**, *1*, 641–656.
- [13] K. Cheng, B. Gu, X. L. Liu, J. C. Kang, Q. H. Zhang, Y. Wang, *Angew. Chem. Int. Ed.* **2016**, *55*, 4725–4728.
- [14] R. A. Buyanov, N. A. Pakhomov, *Kinet. Catal.* **2001**, *42*, 64–75.
- [15] B. Jongsomjit, J. Panpranot, J. G. Goodwin, *J. Catal.* **2001**, *204*, 98–109.
- [16] Y. Xing, Z. Liu, Y. Xue, D. Wu, S. Fang, *Catal. Lett.* **2016**, *146*, 682–691.
- [17] Z. Liu, Y. Xing, Y. Xue, D. Wu, S. Fang, *J. Nanopart. Res.* **2015**, *17*, DOI: 10.1007/s11051-015-2899-3
- [18] Z. Liu, Y. Xue, D. Wu, Y. Xing, S. Fang, *Catal. Lett.* **2015**, *145*, 1941–1947.
- [19] P. Zhai, C. Xu, R. Gao, X. Liu, M. Li, W. Li, X. Fu, C. Jia, J. Xie, M. Zhao, X. Wang, Y.-W. Li, Q. Zhang, X.-D. Wen, D. Ma, *Angew. Chem. Int. Ed.* **2016**, *55*, 9902–9907.
- [20] Z. Pan, D. B. Bukur, *Appl. Catal. A* **2011**, *404*, 74–80.
- [21] X. Wang, W. Ning, L. Hu, Y. Li, *Catal. Commun.* **2012**, *24*, 61–64.
- [22] Z. Pan, M. Parvari, D. B. Bukur, *Appl. Catal. A* **2014**, *480*, 79–85.

- [23] R. P. Marin, S. A. Kondrat, T. E. Davies, D. J. Morgan, D. I. Enache, G. B. Combes, S. H. Taylor, J. K. Bartley, G. J. Hutchings, *Catal. Sci. Technol.* **2014**, *4*, 1970–1978.
- [24] A. D. Fronzo, C. Pirola, A. Comazzi, F. Galli, C. L. Bianchi, A. D. Michele, R. Vivani, M. Nocchetti, M. Bastianini, D. C. Boffito, *Fuel*, **2014**, *119*, 62–69.
- [25] M. Feyzi, M. M. Khodaei, J. Shahmoradi, *Fuel Process. Technol.* **2012**, *93*, 90–98.
- [26] N. N. Madikizela-Mnqanqeni, N. J. Coville, *J. Mol. Catal. A* **2005**, *225*, 137–142.
- [27] S. Li, A. Li, S. Krishnamoorthy, E. Iglesia, *Catal. Lett.* **2001**, *77*, 197–205.
- [28] B. C. Enger, Å.-L. Fossan, Ø. Borg, E. Rytter, A. Holmen, *J. Catal.* **2011**, *284*, 9–22.
- [29] X. Mo, Y.-T. Tsai, J. Gao, D. Mao, J. G. J. Goodwin, *J. Catal.* **2012**, *285*, 208–215.
- [30] Z. Liu, D. Wu, Y. Xing, X. Guo, S. Fang, *Appl. Catal. A-Gen.* **2016**, *514*, 164–172.
- [31] Y. Xing, C. Zhao, G. Jia, S. Fang, Z. Liu, *J. Nanopart. Res.* **2018**, *20*, <https://doi.org/10.1007/s11051-018-4304-5>
- [32] E. Iglesia, S. C. Reyes, R. J. Madon, S. L. Soled, *Advances in Catalysis, Vol 39*, (Eds.: D. D. Eley, H. Pines, P. B. Weisz), Academic Press Inc. **1993**, p. 221–302.
- [33] E. Iglesia, S. C. Reyes, S. L. Soled, *Chemical Industries, Vol 51 (Computer-Aided Design of Catalysts)*, (Eds.: R. E. Becker, C. J. Pereira), Marcel Dekker, New York, **1993**, p. 199–257.
- [34] N. Iyi, S. Takekawa, S. Kimura, *J. Solid State Chem.* **1989**, *83*, 8–19.
- [35] H. K. Kang, H. C. Park, K. H. Kim, *J. Mater. Sci. Mater. Electron.* **1996**, *7*, 385–389.
- [36] Y. Xing, Z. Liu, S. L. Suib, *Chem. Mater.* **2007**, *19*, 4820–4826.
- [37] E. Smit, F. Cinquini, A. M. Beale, O. V. Safonova, W. Beek, P. Sautet, B. M. Weckhuysen, *J. Am. Chem. Soc.* **2010**, *132*, 14928–14941.
- [38] P. Tan, *J. Catal.* **2016**, *338*, 21–29.
- [39] C. S. Kuivila, J. B. Butt, P. C. Stair, *Appl. Surf. Sci.* **1988**, *32*, 99–121.
- [40] T. An, D. Dang, Y. Cheng, *Chem. Res. Chin. Univ.* **2017**, *33*, 672–677.
- [41] A. Loaiza-Gil, B. Fontal, F. Rueda, *Appl. Catal. A* **1999**, *177*, 193–203.
- [42] K. S. W. Sing, D. H. Everett, R. A. W. Haul, L. Moscou, R. A. Pierotti, J. Rouquerol, T. Simmieniowska, *Pure Appl. Chem.* **1985**, *57*, 603–619.
- [43] P. M. Maitlis, V. Zanotti, *Chem. Commun.* **2009**, *0*, 1619–1634.
- [44] N. S. Govender, F. G. Botes, M. H. J. M. de Croon, J. C. Schouten, *J. Catal.* **2014**, *312*, 98–107.
- [45] M. E. Dry, T. Shingles, L. J. Boshoff, G. J. Oosthuizen, *J. Catal.* **1969**, *15*, 190–199.
- [46] M. Ojeda, R. Nabar, A. U. Nilekar, A. Ishikawa, M. Mavrikakis, E. Iglesia, *J. Catal.* **2010**, *272*, 287–297.
- [47] A. K. Dalai, B. H. Davis, *Appl. Catal. A* **2008**, *348*, 1–15.
- [48] X. Chen, D. Deng, X. Pan, Y. Hu, X. Bao, *Chem. Commun.* **2015**, *51*, 217–220.
- [49] M. Luo, B. H. Davis, *Appl. Catal. A* **2003**, *246*, 171–181.

Manuscript received: January 4, 2019
Revised manuscript received: April 16, 2019
Accepted manuscript online: April 29, 2019
Version of record online: May 27, 2019